

AEC HW3

① i)

voltage input $-2 \rightarrow 2$

current output $4\text{mA} \rightarrow 20\text{mA}$

looking at the loop, we have 1 connection from transmitter to ground, so need to use a grounded-load conv.

Consider Howland Pump (one way):

Assuming ideal resistances, $i_o = \frac{1}{R_1} V_i$

then let's assume all resistors are R_1

what does R_1 need to be?

let's just scale ranges:

$$V_{\text{range}} = 2 - (-2) = 4\text{V}$$

$$I_{\text{range}} = 20 - 4 = 16\text{mA}$$

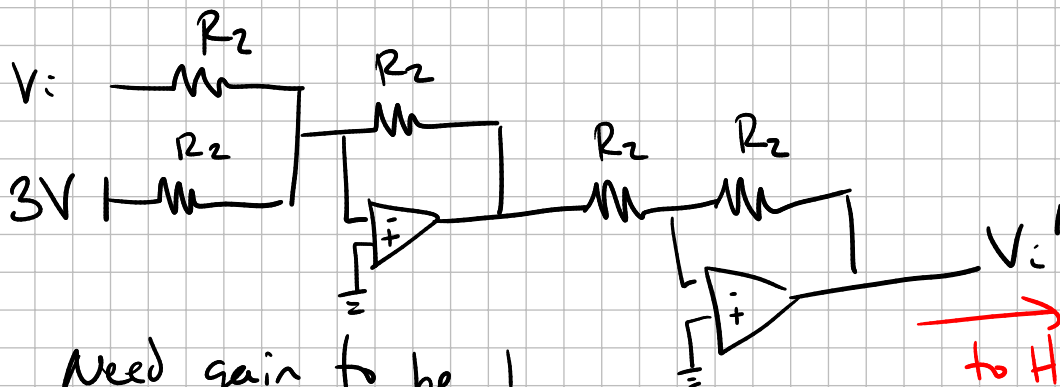
$$I_{\text{range}} = A V_{\text{range}} = \frac{1}{R_1} V_{\text{range}}$$

with this range, what input value corresponds with 4mA ?

$$V_i = 4\text{mA} R_1 = 1\text{V}$$

Need a preamp that shifts input voltage range from $-2 - 2$ to $1 - 5$

Easy enough to do with summing amp & inverter



Need gain to be 1

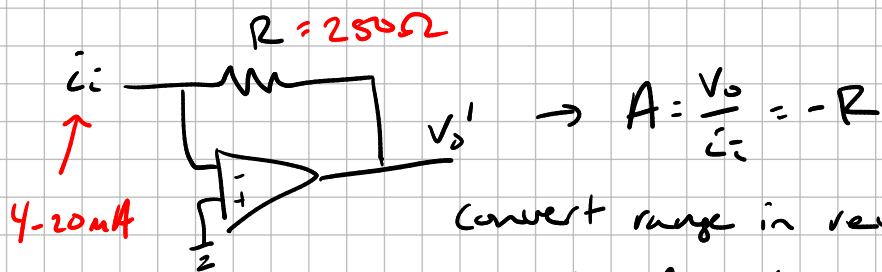
throughout so set all resistors

to same value, pick a reasonable value: $R_2 = 10\text{k}$

to Howland current pump

① ii) Convert 4mA-20mA back to voltage:
0-5V

Basic I-to-V works here:



convert range in reverse:

$$16\text{mA} \rightarrow 4\text{V} \therefore -A = \frac{-4\text{V}}{16\text{mA}} = 250\Omega = R$$

$$V_o' \text{ will be } 4\text{mA} \cdot 250\Omega = -1\text{V}$$

$$16\text{mA} \cdot 250\Omega = -5\text{V}$$

could fix offset with summer (and get desired inversion as well)

Instead, let's try Diff Amp:

$$V_o = A(V_2 - V_1)$$

since need inversion, $V_1 = V_o'$ (because of - sign)

scale needs to be increased from $-1 \rightarrow -5$ to $0-5$
(4) (5)

$$\text{therefore: } A = \frac{5}{4} = 1.25$$

then calculate V_2 :

$$0\text{V} = 1.25(V_2 - -1), V_2 = -1\text{V}$$

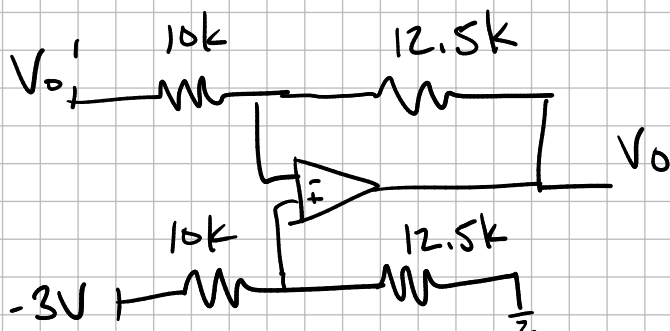
check:

$$5\text{V} = 1.25(-1\text{V} - -5) = 5\text{V} \checkmark$$

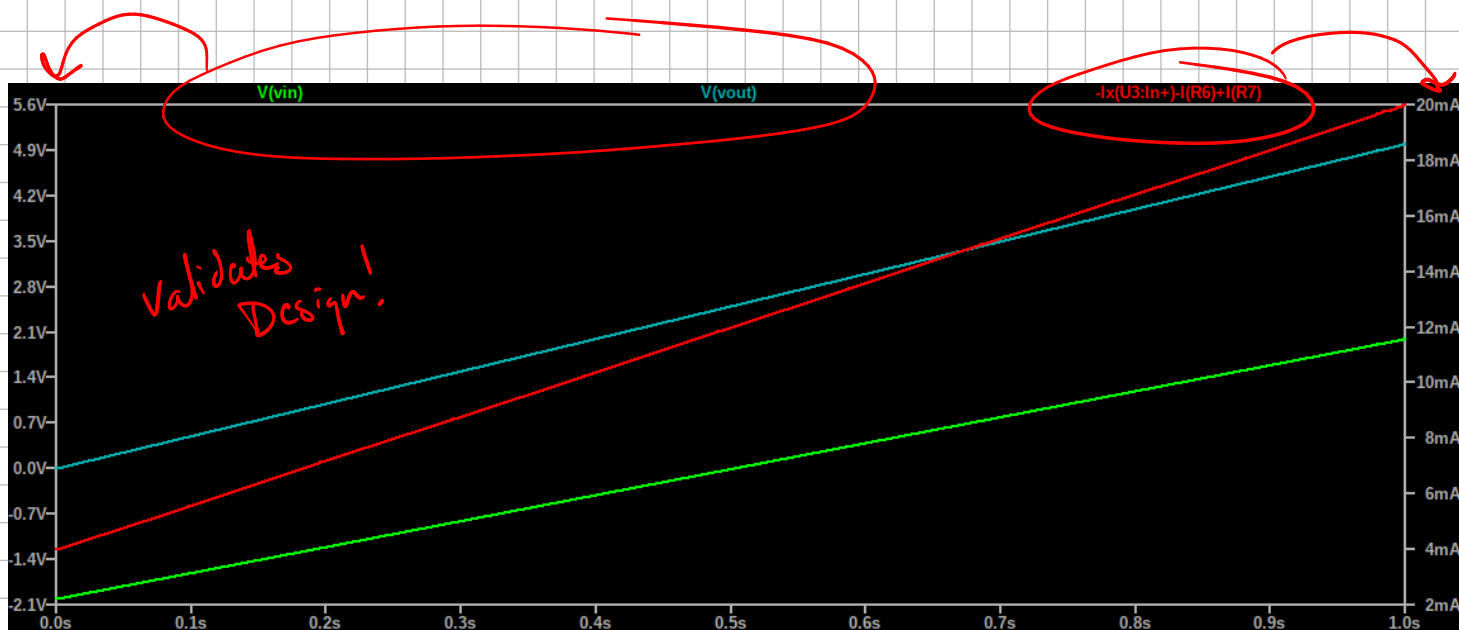
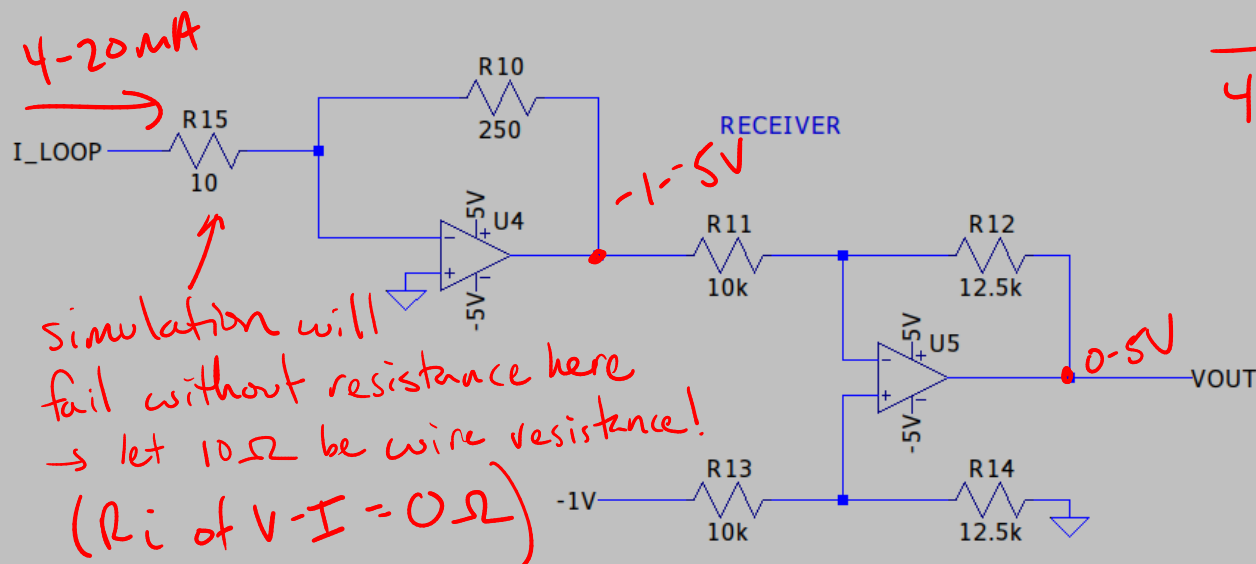
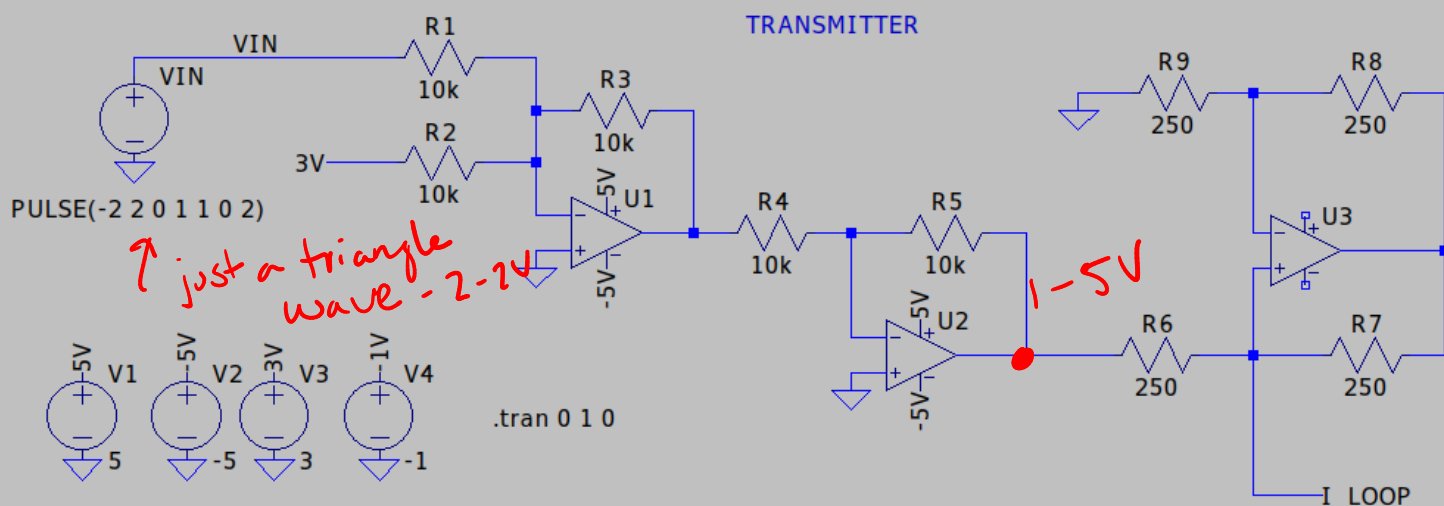
for Diff Amp, $A = \frac{R_2}{R_1}$, $R_2 = 1.25R_1$

let $R_1 = 10\text{k}$

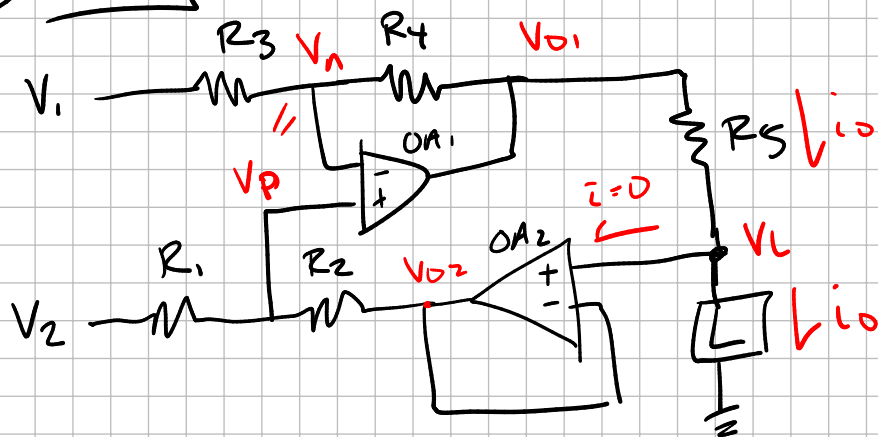
$$\therefore R_2 = 12.5\text{k}$$



iii)



② 2.21



$V_{o2} = V_L$, OA2 is Voltage follower

$$V_p = (V_L - V_2) \frac{R_1}{R_1 + R_2} + V_2 \quad \leftarrow \text{voltage divider for } V_p$$

$$V_n = (V_{o1} - V_1) \frac{R_3}{R_3 + R_4} + V_1 \quad \leftarrow \text{voltage divider for } V_n$$

Virtual short: $V_p = V_n$

$$(V_L - V_2) \frac{R_1}{R_1 + R_2} + V_2 = (V_{o1} - V_1) \frac{R_3}{R_3 + R_4} + V_1$$

$$V_{o1} - V_L = i_o R_5 \quad \leftarrow \text{Ohm's law across } R_5$$

$$V_{o1} = i_o R_5 + V_L \quad \text{substitute into } \star:$$

$$(V_L - V_2) \frac{R_1}{R_1 + R_2} + V_2 = (i_o R_5 + V_L - V_1) \frac{R_3}{R_3 + R_4} + V_1$$

$$\frac{R_1}{R_1 + R_2} V_L + V_2 \left(1 - \frac{R_1}{R_1 + R_2}\right) = \left(i_o + \frac{V_L}{R_5}\right) \frac{R_3 R_5}{R_3 + R_4} + V_1 \left(1 - \frac{R_3}{R_3 + R_4}\right)$$

$$i_o = \frac{R_3 + R_4}{R_3 R_5} \left[V_2 \left(1 - \frac{R_1}{R_1 + R_2}\right) - V_1 \left(1 - \frac{R_3}{R_3 + R_4}\right) - V_L \left(\frac{R_3}{R_3 + R_4} - \frac{R_1}{R_1 + R_2}\right) \right]$$

$$A_1 = \frac{R_3 + R_4}{R_3 R_5} \left(1 - \frac{R_3}{R_3 + R_4}\right) = \frac{R_4}{R_3 R_5}$$

$$A_2 = \frac{R_3 + R_4}{R_3 R_5} \left(1 - \frac{R_1}{R_1 + R_2}\right) = \left(\frac{R_3 + R_4}{R_3}\right) \left(\frac{R_2}{R_1 + R_2}\right) \frac{1}{R_5}$$

$$R_o = \frac{R_3 R_5}{R_3 + R_4} / \left(\frac{R_1}{R_3 + R_4} - \frac{R_1}{R_1 + R_2}\right)$$

$$= (R_1 + R_2) R_3 R_5 / (R_1 R_3 + R_2 R_3 - R_1 R_3 + R_1 R_4)$$

$$= (R_1 + R_2) R_3 R_5 / (R_2 R_3 - R_1 R_4)$$

$$= \left(1 + \frac{R_1}{R_2}\right) R_5 / \left(1 - \frac{R_1 R_4}{R_2 R_3}\right)$$

$$R_o = \left(\frac{R_2}{R_1} + 1\right) R_5 / \left(\frac{R_2}{R_1} - \frac{R_4}{R_3}\right)$$

for $R_o \rightarrow \infty$, $\frac{R_2}{R_1} = \frac{R_4}{R_3}$

if this is the case, then

$$A_1 = \frac{R_2}{R_1 R_5}, \quad A_2 = \left(\frac{R_1 + R_2}{R_1}\right) \left(\frac{R_2}{R_1 + R_2}\right) \frac{1}{R_5}$$

$$A_2 = \frac{R_2}{R_1 R_5}$$

$$A_1 = A_2$$

$$i_o = A(V_2 - V_1) \rightarrow A = \frac{R_2}{R_1 R_5}$$

GRAD ONLY

2.21b) for R_0 , we can get to ∞ by forcing the denominator to 0.

$$\rightarrow \frac{R_2}{R_1} - \frac{R_4}{R_3} = 0 \rightarrow \frac{R_2}{R_1} = \frac{R_4}{R_3}$$

worst case scenario for 1% resistors is:

$$\frac{R_2(1+1\%)}{R_1(1-1\%)} = \frac{R_4(1-1\%)}{R_3(1+1\%)} \Rightarrow \frac{R_2}{R_1} = \frac{R_4}{R_3} \frac{(1-1\%)^2}{(1+1\%)^2} = \frac{R_4}{R_3} \underbrace{(1-1\%)^4}$$

$$(1-p)^4 = 1 - 4p + 6p^2 - 4p^3 + p^4$$

since $p \ll 1$, then $4p \gg 6p^2 \gg 4p^3 \gg p^4$

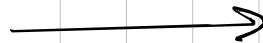
$$(1-p)^4 \approx 1 - 4p$$

$$\frac{R_2}{R_1} = \frac{R_4}{R_3} (1-\varepsilon) \quad \text{where} \quad \varepsilon \approx 4p = 0.04$$

with error, R_0 becomes:

$$R_0 = \left(\frac{R_2}{R_1} + 1\right) R_5 / \left(\frac{R_2}{R_1} - \frac{R_2}{R_1} (1-\varepsilon)\right) = \left(\frac{R_2}{R_1} + 1\right) R_5 / \frac{R_2}{R_1} \varepsilon$$

$$R_0 = \left(1 + \frac{R_1}{R_2}\right) R_5 / \varepsilon \geq 25 \left(1 + \frac{R_1}{R_2}\right) R_5$$



Next page Not necessary for solution
Done for completeness sake.

similarly, effects on A_1 & A_2 :

$$A_1 = \frac{R_4}{R_3 R_5} = \frac{R_2}{R_1 R_5} (1 - \epsilon)$$

$$A_2 = \left(\frac{R_3 + R_4}{R_3} \right) \left(\frac{R_2}{R_1 + R_2} \right) \frac{1}{R_5} = \left(\frac{R_1 + R_2 (1 - \epsilon)}{R_1} \right) \left(\frac{R_2}{R_1 + R_2} \right) \frac{1}{R_5}$$

$$A_2 = \frac{R_2}{R_1 R_5} \left(\frac{R_1 + R_2 (1 - \epsilon)}{R_1 + R_2} \right)$$

$$i_o = A_2 v_2 - A_1 v_1 - v_L / R_{L0}$$

← focus on that

$$\frac{R_2}{R_1 R_5} \left[\frac{R_1 + R_2 (1 - \epsilon)}{R_1 + R_2} v_2 - (1 - \epsilon) v_1 \right]$$

Sub in v_{cm} , v_{dm} :

$$\frac{R_2}{R_1 R_5} \left[\frac{R_1 + R_2 (1 - \epsilon)}{R_1 + R_2} \left(v_{cm} + \frac{v_{dm}}{2} \right) - (1 - \epsilon) \left(v_{cm} - \frac{v_{dm}}{2} \right) \right]$$

Then

$$A_{cm} = \frac{R_2}{R_1 R_5} \left[\frac{R_1 + R_2 (1 - \epsilon)}{R_1 + R_2} - \frac{(R_1 + R_2) (1 - \epsilon)}{R_1 + R_2} \right]$$

$$A_{cm} = \frac{\epsilon R_2}{R_5 (R_1 + R_2)}$$

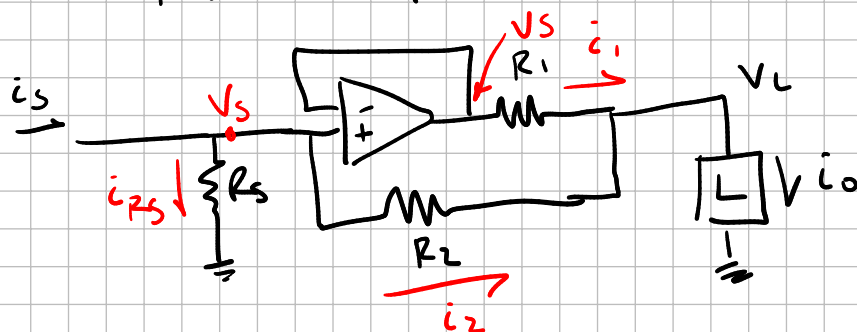
$$A_{dm} = \frac{1}{2} \frac{R_2}{R_1 R_5} \left[\frac{R_1 + R_2 (1 - \epsilon)}{R_1 + R_2} + \frac{(R_1 + R_2) (1 - \epsilon)}{R_1 + R_2} \right]$$

$$A_{dm} = \frac{1}{2} \frac{R_2}{R_1 R_5 (R_1 + R_2)} \left[R_1 (2 - \epsilon) + 2 R_2 (1 - \epsilon) \right]$$

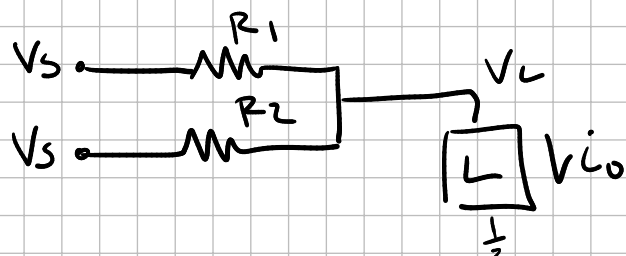
$$A_{dm} \approx R_2 / R_1 R_5 \quad CMRR \approx 20 \log_{10} \left[\epsilon R_1 / R_1 + R_2 \right]$$

2.25

find R_o & A :



using virtual short, we have



$$i_o = \frac{V_s - V_L}{R_1 \parallel R_2} = (V_s - V_L) \frac{R_1 + R_2}{R_1 R_2}$$

current balance on input:

$$i_s = i_{R_s} + i_2 = \frac{V_s}{R_s} + \frac{V_s - V_L}{R_2}$$

$$V_s = \frac{i_s + V_L/R_2}{\frac{1}{R_s} + \frac{1}{R_2}} = R_s \parallel R_2 (i_s + V_L/R_2)$$

$$= \frac{R_s R_2}{R_s + R_2} (i_s + V_L/R_2)$$

$$i_o = \frac{R_1 + R_2}{R_1 R_2} \left[\frac{R_s R_2}{R_s + R_2} (i_s + V_L/R_2) - V_L \right]$$

$$= \frac{R_s}{R_1} \frac{R_1 + R_2}{R_s + R_2} i_s - \frac{R_1 + R_2}{R_1 R_2} V_L \left(1 - \frac{R_s}{R_s + R_2} \right)$$

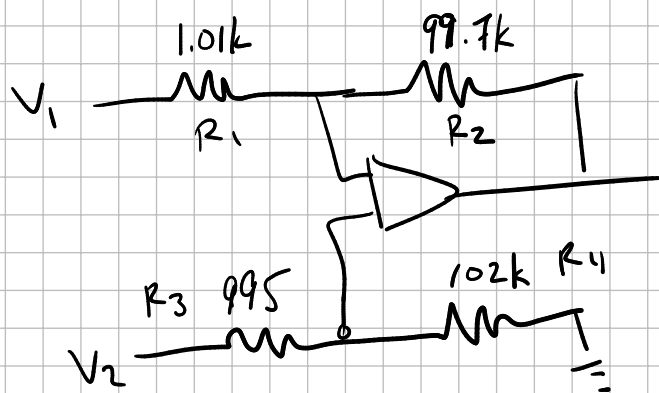
$$A = \frac{R_s}{R_1} \frac{R_1 + R_2}{R_s + R_2} = \frac{1 + R_2/R_1}{1 + R_2/R_s}$$

$$R_o = \frac{R_1 R_2}{R_1 + R_2} \left/ 1 - \frac{R_s}{R_s + R_2} \right. = \frac{R_1 R_2}{R_1 + R_2} \left/ \frac{R_2}{R_s + R_2} \right.$$

$$R_o = \frac{R_1 (R_s + R_2)}{R_1 + R_2} = \frac{R_s + R_2}{1 + R_2/R_1}$$

2.34

Standard Diff Amp.



find A_{dm} , A_{cm} , $CMRR_{dB}$

Actual values:

$$V_0 = \frac{R_4}{R_1} \left(\frac{R_1 + R_2}{R_3 + R_4} \right) V_2 - \frac{R_2}{R_1} V_1 = 98.7496 V_2 - 98.7129 V_1$$

$$= 98.7496 \left(V_{cm} + \frac{V_{dm}}{2} \right) - 98.7129 \left(V_{cm} - \frac{V_{dm}}{2} \right)$$

$$= 0.0367 V_{cm} + 98.7312 V_{dm}$$

$$A_{cm} = 0.0367 \text{ V/V}$$

$$A_{dm} = 98.7312 \text{ V/V}$$

$$CMRR_{dB} = 20 \log_{10} \left(\frac{A_{dm}}{A_{cm}} \right) = 68.6 \text{ dB}$$

Estimated:

$$\frac{R_3}{R_4} = \frac{R_1}{R_2} (1 - \epsilon) \quad 1 - \epsilon = 0.9629 \rightarrow \epsilon = 0.0371$$

$$A_{cm} = \frac{R_2}{R_1 + R_2} \epsilon = 0.0367 \text{ V/V}$$

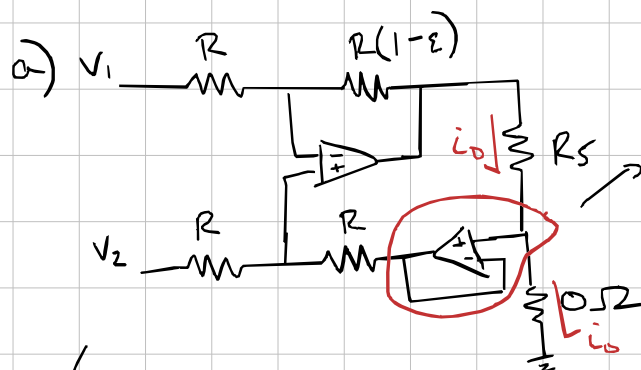
$$A_{dm} \approx \frac{R_2}{R_1} = \frac{100}{1} = 100 \text{ V/V}$$

R_2, R_1 ideal values

$$CMRR_{dB} = 20 \log_{10} \left(\frac{100}{0.0367} \right) = 68.7 \text{ dB}$$

GRAD ONLY

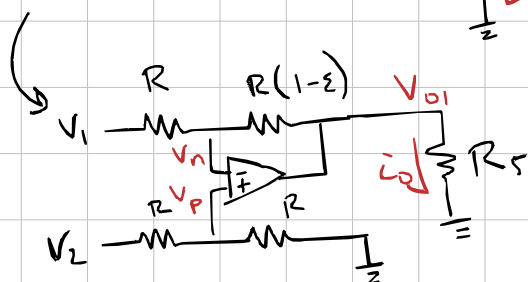
2.39 | Since A_{un} & A_{om} defined when output load is shorted, analyze problem with $V_L = 0V$:



Non-ideal buffer:

$$V_o = a(V_p - V_n)$$

$$V_p = 0V \therefore V_o = 0V$$



Just a difference amp with R_S as a load!

$$i_o = V_{o1} / R_S$$

via superposition:

A) $V_1 \neq 0, V_2 = 0$: $V_p = 0, V_{o1} = a(0 - V_n)$

$$V_n = (V_{o1} - V_1) \frac{R}{R + R(1-\epsilon)} + V_1 = -\frac{V_{o1}}{a}$$

$$-\left(\frac{1}{a} + \frac{1}{2-\epsilon}\right) V_{o1} = \left(\frac{-1}{2-\epsilon} + \frac{2-\epsilon}{2-\epsilon}\right) V_1$$

$$V_{o1} = -(1-\epsilon) / \left(\frac{2-\epsilon}{a} + 1\right) V_1$$

B) $V_1 = 0, V_2 \neq 0$: $V_p = V_2/2, V_{o1} = a(V_p - V_n) \rightarrow V_n = \frac{V_2}{2} - \frac{V_{o1}}{a}$

$$V_n = V_{o1} \frac{R}{R + R(1-\epsilon)} = V_{o1} / (2-\epsilon) = \frac{V_2}{2} - \frac{V_{o1}}{a}$$

$$V_{o1} = (2-\epsilon) / \left(\frac{2-\epsilon}{a} + 1\right) (V_2/2) = (1-\epsilon/2) / \left(\frac{2-\epsilon}{a} + 1\right) V_2$$

A) + B):

$$V_{o1} = \frac{1}{\left(\frac{2-\epsilon}{a} + 1\right)} \left[(1-\epsilon/2) V_2 - (1-\epsilon) V_1 \right]$$

$$V_{O1} = \frac{1}{\left(\frac{2-\varepsilon}{a} + 1\right)} \left[\left(1 - \frac{\varepsilon}{2}\right) V_2 - (1 - \varepsilon) V_1 \right]$$

with $V_1 = V_{cm} - V_{dm}/2$, $V_2 = V_{cm} + V_{dm}/2$
common mode:

$$V_{O1(cm)} = \frac{1}{\left(\frac{2-\varepsilon}{a} + 1\right)} \left[\left(1 - \frac{\varepsilon}{2}\right) V_{cm} - (1 - \varepsilon) V_{cm} \right]$$

$$V_{O1(cm)} = \frac{1}{\left(\frac{2-\varepsilon}{a} + 1\right)} \left[\frac{\varepsilon}{2} V_{cm} \right]$$

$$\bar{I}_{cm} = \left(\frac{1}{\frac{2-\varepsilon}{a} + 1} \right) \left[\frac{\varepsilon}{2} V_{cm} \right] / R_5, \quad A_{cm} = \frac{1}{\frac{2-\varepsilon}{a} + 1} \left(\frac{\varepsilon}{2} \right) \frac{1}{R_5}$$

Diff mode:

$$V_{O1(dm)} = \frac{1}{\left(\frac{2-\varepsilon}{a} + 1\right)} \left[\left(1 - \frac{\varepsilon}{2}\right) \frac{V_{dm}}{2} + (1 - \varepsilon) \frac{V_{dm}}{2} \right]$$

$$V_{O1(dm)} = \frac{1}{\left(\frac{2-\varepsilon}{a} + 1\right)} \left[\left(1 - \frac{3\varepsilon}{4}\right) V_{dm} \right] \quad \text{since } \varepsilon \ll 1, \text{ then:}$$

$$V_{O1(dm)} \approx \frac{1}{\left(\frac{2-\varepsilon}{a} + 1\right)} V_{dm}, \quad \bar{I}_{dm} = \frac{1}{\left(\frac{2-\varepsilon}{a} + 1\right)} V_{dm} / R_5$$

$$A_{dm} = \frac{1}{\frac{2-\varepsilon}{a} + 1} \left(\frac{1}{R_5} \right)$$

$$CMRR = 20 \log_{10} \left(\frac{A_{dm}}{A_{cm}} \right) = 20 \log_{10} \left(\frac{2}{\varepsilon} \right)$$

with R tolerances of $\pm 1\%$, then $\rho = 1\%$, $\varepsilon = 0.04$
(see ex. 2.9)

$$CMRR = 20 \log_{10} \left(\frac{2}{0.04} \right) = 34 \text{ dB}$$

Note: CMRR is agnostic of a ! $\therefore$ Part B:
 $CMRR \rightarrow \infty$